



PROJECT 10

MARKETING ANALYTICS

Model Evaluation with Cross Validation

PRESENTED BY MUNAWAR

TOPIC REVIEW - CROSS VALIDATION

Model Evaluation is the process of assessing how well a model (such as a regression or classification model) performs in making predictions or decisions based on data. Why Model Evaluation is Important

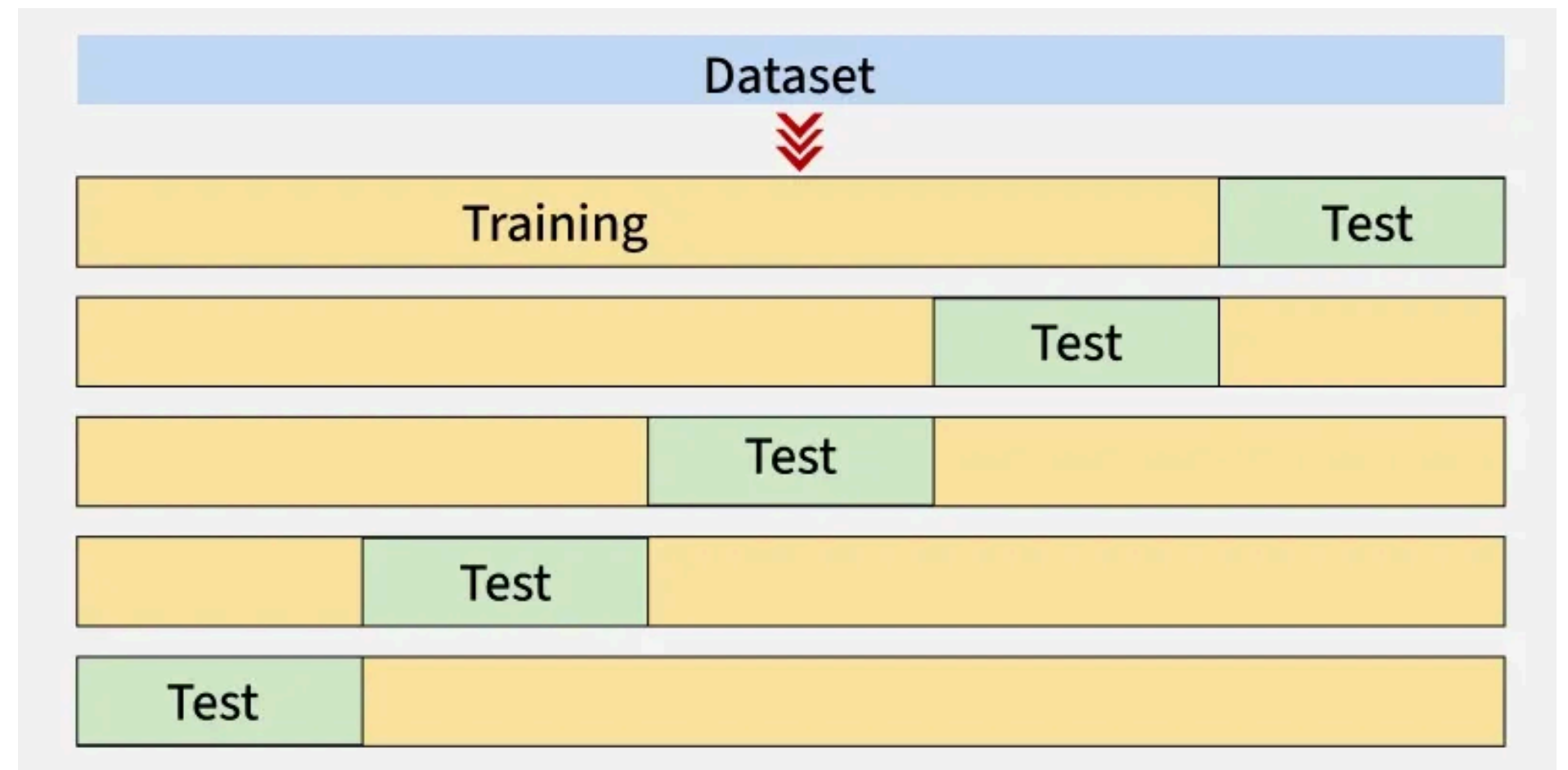
- Avoiding Overfitting and Underfitting
- Choosing the Best Model
- Measuring Business Impact

In this project, I used k-fold cross validation in conducting model evaluation. K-Fold Cross Validation is a validation technique used to test model performance by dividing the dataset into several subsets, called folds. This technique is designed to address the problem of overfitting, where a model performs very well on training data but poorly on predicting new data. By using k-fold cross-validation, we can obtain more stable and representative model evaluation results.

CROSS VALIDATION PROCESS

The process can be broken into the following steps:

1. Split Data into K Folds: Divide the dataset into K subsets or folds.
2. Train-Test Iterations: For each fold:
 - Use $K - 1$ folds for training the model.
 - Use the remaining fold as the test set to evaluate the model.
3. Aggregate Results: Calculate the performance metric (e.g., accuracy, precision, recall, etc.) for each fold and average the results.



BACKGROUND

In a fast-paced industry, relying solely on intuition is no longer enough to determine production volumes and inventory management. By leveraging historical marketing performance data—from ad clicks to social media views to direct consumer interactions—we can understand potential buyers' behavior patterns before they make a purchase.

Through this analytical approach, the sales team isn't just guessing numbers; it's scientifically predicting market demand. As a result, we can optimize inventory, reduce holding costs, and ensure that every dollar invested in digital marketing directly correlates to increased sales volume.

BUSINESS QUESTIONS

Question 1

How much does advertising influence purchases?

- Identify the key drivers of social purchases
- Quantify the effect of social clicks on conversions
- Understand category-level impact (Base, Elite, Game)

Question 2

If the sales target is 150, how many clicks are needed?

- Use regression equation to back-solve for clicks
- Calculate per product category (Base, Elite, Game)
- Recommend most efficient category for campaigns

MODEL & METHODOLOGY

$$\text{socialpurchase} = -2.19 + (0.0655 \times \text{socialclicks}) + (10.61 \times 1_{\text{elite}}) + (-5.99 \times 1_{\text{game}})$$

social_clicks

+0.0655

Every 15-16 clicks → +1 purchase

Elite category

+10.61

vs Base at same click level

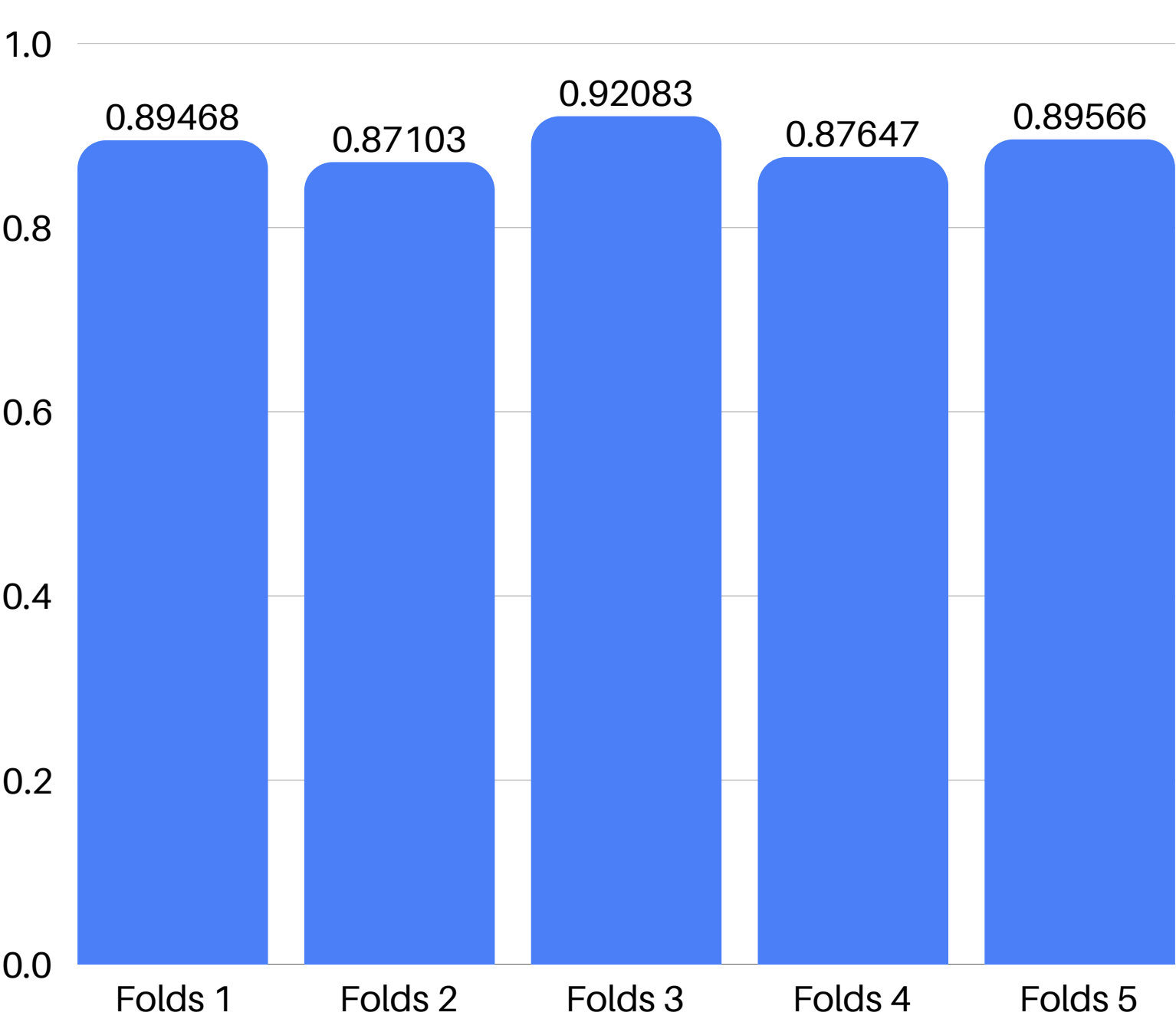
Game category

-5.99

vs Base at same click level

- Full model $R^2 = 0.9013$ (90.1% variance explained)
- meaning the model is able to explain **90.13% of the variance** in *social_purchases*.

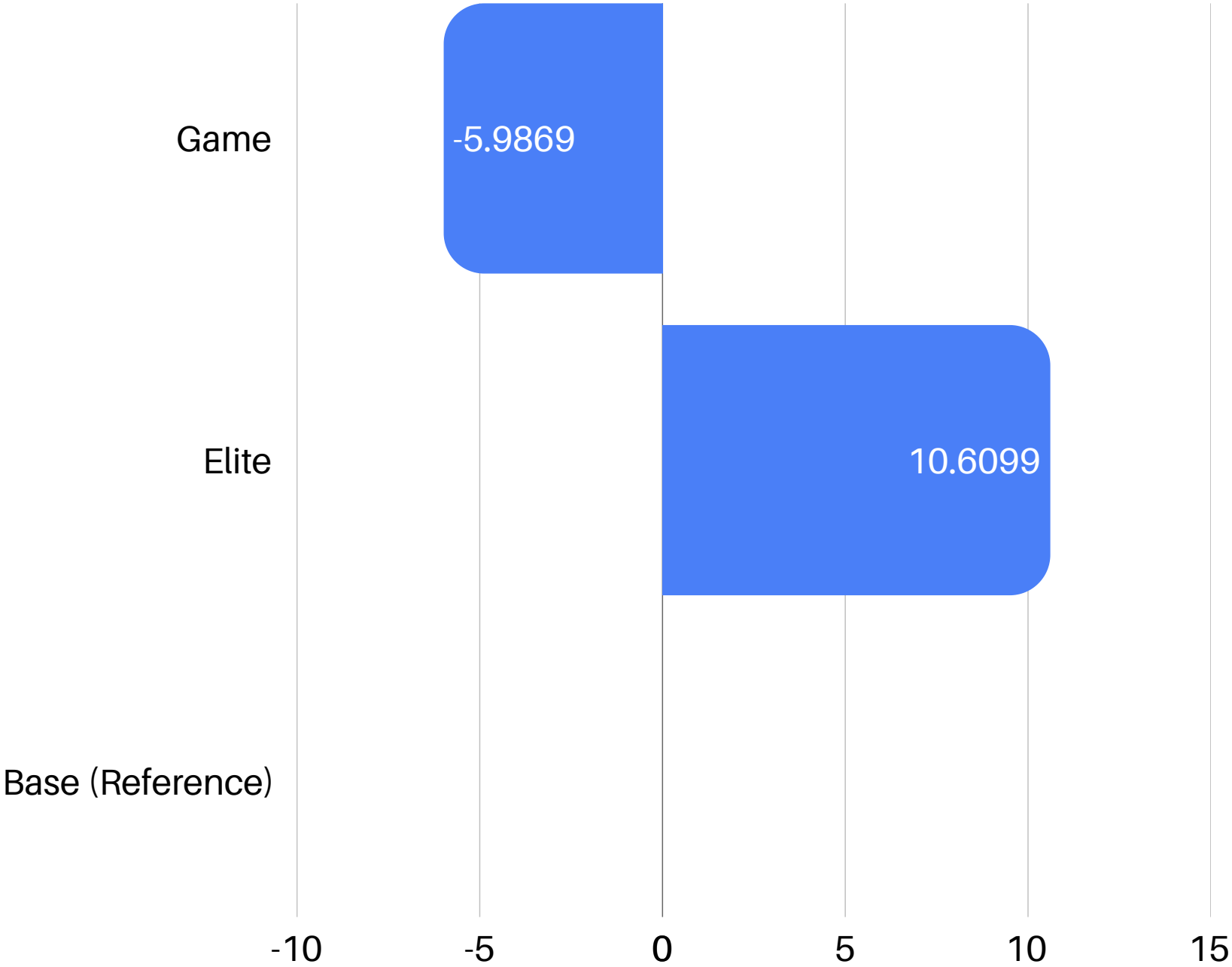
MODEL VALIDATION – 5-FOLD CROSS-VALIDATION



Summary

Mean R ²	→	0.8917
Best Fold	→	Fold 3
Best R ²	→	0.9208
Min R ²	→	0.8710
Std Dev	→	0.019

CATEGORY IMPACT ON PURCHASES



Elite **+10.61**

10.61 more purchases than Base at same clicks.
Highest conversion rate.

Game **-5.99**

5.99 fewer purchases than Base. Review ad
content & landing pages.

MODEL & METHODOLOGY

$$\begin{aligned} \textit{socialclicks} &= (150 - \textit{intercept} - \textit{wcategory}) \div \textit{wclicks} \\ &150 - (-2.19) - \textit{wcat}) \div 0.0655 \end{aligned}$$

Base 2,323 Reference category	Elite 2,161 Most efficient! 162 fewer clicks	Game 2,414 91 more clicks than Base
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CONCLUSION

1

Social clicks drive purchases

Every 15–16 additional social clicks produces 1 additional purchase.

2

Category matters significantly

Elite generates 10.61 more sales than Base; Game underperforms by 5.99.

3

Model is reliable

CV $R^2 = 0.892$, Full $R^2 = 0.901$ — stable across all validation folds.

4

Strategic Recommendation

Prioritize Elite category campaigns to maximize conversion efficiency. Aim for ~2,161 social clicks to achieve the 150-purchase target.



THANK YOU FOR YOUR TIME



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DATA ANALYST PRESENTATION
